

Missing Features — Linear Task Backlog

Current state summary

Pleros is a **feature-rich SMB wholesale ERP** (admin + B2B shop + 3 mobile PWAs) in [apps/client](#) with Express/Prisma API in [apps/web/lib/server](#). Core workflows (order-to-cash, WMS, finance, EDI, MSA) are largely shipped per [ERP_FEATURE_GAP.md](#).

What you asked about:

Area	Reality today	Evidence
Age verification	Not implemented — only metadata (tobacco SKU flag, customer license field, onboarding checkbox)	inventory/schema.prisma <code>isTobacco</code> ; crm/schema.prisma <code>tobaccoLicenseNumber</code> ; no gates in POS/checkout
Label scanner	Text input only — no camera/OCR	m/warehouse/receiving/page.tsx L66–74
PWA	Basic — manifest + SW registered, but incomplete offline/install story	manifest.webmanifest , sw.js , main.tsx L16–19

Diagram omitted in PDF — see the plan markdown for the mermaid flowchart.

Epic 1: Age Verification & Regulated Product Compliance

Context: Vertical enums (`TOBACCO_VAPE` , `ALCOHOL`) and tobacco metadata exist but **nothing enforces age or license at sale**. Onboarding [COMPLIANCE_ACK](#) is an admin acknowledgment, not a buyer/POS age gate.

COS-101 — Define age verification policy model

- **Priority:** High | **Labels:** compliance, backend, schema
- **Description:** Add tenant-level config for regulated-product rules: minimum age (default 21), required customer license types, which surfaces require verification (POS, B2B checkout, delivery POD).
- **Acceptance criteria:**
 - New fields on tenant settings (e.g. `ageVerificationEnabled` , `minimumAge` , `requireTobaccoLicense`)
 - Prisma migration in [tenant/schema.prisma](#)
 - Settings UI in [admin/settings/page.tsx](#)
 - API: `GET/PATCH /tenants/me/compliance-settings`

COS-102 — Enforce licensed-customer rules for tobacco SKUs (B2B)

- **Priority:** High | **Labels:** compliance, shop, backend
- **Description:** Block or warn when unlicensed buyers add `isTobacco` SKUs to cart/checkout.
- **Acceptance criteria:**
 - Server-side validation in order confirm / checkout ([orders.ts](#))
 - Catalog/cart UI shows regulated-product badge and license requirement
 - Error message when `customer.isLicensedTobacco` is false or `tobaccoLicenseNumber` missing/expired

COS-103 — POS age verification gate

- **Priority:** High | **Labels:** compliance, pos, frontend
- **Description:** When cart contains regulated SKUs, require staff to confirm customer age/license before completing sale at [admin/pos/page.tsx](#).
- **Acceptance criteria:**
 - Modal/step: DOB entry or "ID verified" attestation with staff user logged
 - Audit log entry for each verification ([audit-log.ts](#))
 - Sale blocked until verified when tenant policy enabled

COS-104 — Age-restricted SKU flag generalization

- **Priority:** Medium | **Labels:** compliance, inventory
- **Description:** Extend beyond `isTobacco` to `ageRestricted` + `minimumAge` per SKU (covers alcohol, vape, pharma).

- **Acceptance criteria:**
 - Schema + inventory create/edit UI updates ([admin/inventory/page.tsx](#))
 - Backward compatible: `isTobacco` maps to `ageRestricted` + age 21

COS-105 — Delivery POD age check for regulated shipments

- **Priority:** Medium | **Labels:** compliance, mobile, dispatch
- **Description:** Driver must confirm recipient age/license before marking regulated stops delivered.
- **Acceptance criteria:**
 - Check order lines for age-restricted SKUs
 - POD form requires age confirmation checkbox or license photo capture
 - Stored on POD record in dispatch schema

COS-106 — (Future) ID document scan integration

- **Priority:** Low | **Labels:** compliance, mobile, integration
- **Description:** Optional integration with ID scanning provider (e.g. Stripe Identity, Onfido, or manual camera capture of license).
- **Acceptance criteria:** Spike doc + provider selection; not required for MVP of COS-101–103

Epic 2: Camera Barcode / Label Scanner

Context: Landing page claims "Barcode scans" ([home/page.tsx](#) L71) but mobile receiving uses a plain text field. Server already resolves scans via [findSkuByScanValue](#) . Permissions-Policy: `camera=(self)` is set in [client/server/index.ts](#) but unused.

COS-201 — Reusable camera barcode scanner component

- **Priority:** High | **Labels:** mobile, frontend, wms
- **Description:** Build `BarcodeScanner` component using `BarcodeDetector` API with `@zxing/library` fallback for Safari.
- **Acceptance criteria:**
 - Start/stop camera, torch toggle (where supported), manual entry fallback
 - Debounced scan callback with duplicate suppression
 - Works on HTTPS and localhost; graceful degradation message when camera denied

COS-202 — Wire scanner into warehouse receiving

- **Priority:** High | **Labels:** mobile, wms
- **Description:** Replace text-only input in [m/warehouse/receiving/page.tsx](#) with scanner component.
- **Acceptance criteria:** Scan triggers same `POST .../scan` flow; offline queue still works (COS-301)

COS-203 — Wire scanner into pick tasks and wave picking

- **Priority:** Medium | **Labels:** mobile, wms
- **Description:** Add scan-to-confirm on pick lines in [m/warehouse/tasks](#) pages.
- **Acceptance criteria:** Wrong barcode shows error; correct scan advances/confirms pick line

COS-204 — Wire scanner into cycle count (admin + mobile)

- **Priority:** Medium | **Labels:** wms, admin
- **Description:** Scan SKU/bin during cycle count entry.
- **Acceptance criteria:** Scan populates SKU field on admin cycle count and mobile count flows

COS-205 — POS barcode scan input

- **Priority:** Medium | **Labels:** pos, admin
- **Description:** Support USB wedge + optional camera scan on POS register page.
- **Acceptance criteria:** Scan adds SKU to POS cart via existing SKU lookup API

COS-206 — Printable label format improvements

- **Priority:** Low | **Labels:** wms, backend
- **Description:** Enhance [barcode-labels.ts](#) — Code128/QR render, batch print, standard label sizes.
- **Acceptance criteria:** Labels scannable by COS-201 component at 4–6 inch distance

Epic 3: PWA & Offline Mobile Hardening

Context: PWA shell exists but is **not production-grade**. Critical gaps:

- No `registration.sync.register('pleros-offline-queue')` — Background Sync documented in [docs/celestial/06-faq.md](#) L167 but unwired
- Offline queue only replays receiving actions ([offline-sync.ts](#) L5–21)
- SW cache is minimal network-first fallback ([sw.js](#) L11–18)
- Manifest `start_url` is warehouse-only; no install prompt UX

COS-301 — Register Background Sync for offline queue

- **Priority:** High | **Labels:** pwa, mobile
- **Description:** After enqueueing offline actions, call `SyncManager.register('pleros-offline-queue')` and handle permission/feature detection.
- **Acceptance criteria:**
 - [offline-queue.ts](#) triggers sync registration
 - SW `sync` event replays queue via existing `PLEROS_SYNC_QUEUE` message path
 - Update FAQ/docs to match actual behavior

COS-302 — Expand offline action types

- **Priority:** High | **Labels:** pwa, mobile
- **Description:** Queue pick confirmations, POD submissions, sales activity logs offline.
- **Acceptance criteria:**
 - New action types in `offline-sync.ts` replay handlers
 - Mobile pages enqueue when `!navigator.onLine`
 - Conflict handling: show failed count in [OfflineBanner](#)

COS-303 — App-shell caching strategy for mobile routes

- **Priority:** Medium | **Labels:** pwa
- **Description:** Precache `/m/*` static assets and shell; stale-while-revalidate for API GETs used on mobile home.
- **Acceptance criteria:** Mobile warehouse/delivery/sales load with no network after first visit
- **Note:** Consider adopting `vite-plugin-pwa` vs extending hand-rolled [sw.js](#)

COS-304 — PWA install prompt and per-app manifests

- **Priority:** Medium | **Labels:** pwa, mobile
- **Description:** Separate manifest shortcuts or dynamic manifest for Warehouse / Delivery / Sales; "Add to Home Screen" prompt on mobile login.
- **Acceptance criteria:**
 - `beforeinstallprompt` handler on mobile layout
 - Icons/maskable assets verified (512px exists)
 - Each app deep-links to correct `start_url`

COS-305 — Offline conflict resolution UX

- **Priority:** Medium | **Labels:** pwa, mobile
- **Description:** When replay fails (409/stale session), surface actionable errors instead of silent fail.
- **Acceptance criteria:** Banner shows failed action count + link to retry/discard

COS-306 — PWA audit (Lighthouse + iOS/Android)

- **Priority:** Low | **Labels:** pwa, qa
- **Description:** Run Lighthouse PWA audit; fix installability, maskable icon, theme-color meta gaps.
- **Acceptance criteria:** PWA score ≥ 90 on mobile routes

Epic 4: Mobile Field Gaps (marketing vs code)

Context: Landing page lists "POD photos" ([home/page.tsx](#) L80) but delivery sends `photoUrl: null` ([m/delivery/route/fid/page.tsx](#) L42).

COS-401 — POD photo capture and upload

- **Priority:** High | **Labels:** mobile, dispatch

- **Description:** Camera capture on delivery stop completion; upload to S3/local storage; attach URL to POD API.
- **Acceptance criteria:**
 - `<input type="file" capture="environment">` or camera component
 - Backend stores `photoUrl` on POD record
 - Admin dispatch view shows POD photo thumbnail

COS-402 — POD signature capture

- **Priority:** Medium | **Labels:** mobile, dispatch
- **Description:** Canvas signature pad; send `signatureDataUrl` (currently always null).
- **Acceptance criteria:** Signature rendered on admin stop detail

COS-403 — Route stop reorder / basic optimization

- **Priority:** Low | **Labels:** dispatch
- **Description:** Manual drag reorder exists partially; add map view or simple nearest-neighbor sort.
- **Acceptance criteria:** Documented in [ERP_FEATURE_GAP.md](#) L142 as partial

Epic 5: Platform Integration Stubs

Context: Infrastructure exists but is not wired to real events.

COS-501 — Wire webhook auto-dispatch on domain events

- **Priority:** High | **Labels:** integrations, backend
- **Description:** [webhooks.ts](#) only supports manual `testWebhook`. Emit subscribed events (order.shipped, compliance.batch_recalled, etc.) from order/compliance flows.
- **Acceptance criteria:**
 - Dispatcher called from order saga, invoice issue, MSA submit, batch recall
 - Events match list in [webhook-manager.tsx](#)
 - HMAC signature + SSRF checks preserved

COS-502 — Implement Redis event bus

- **Priority:** High | **Labels:** infra, backend
- **Description:** Replace console stub in [event-bus.ts](#) with ioredis pub/sub; call `publishOrderEvent` from order lifecycle.
- **Acceptance criteria:** Events published when `REDIS_URL` set; falls back to console in dev

COS-503 — Redis-backed rate limiting for multi-replica

- **Priority:** Medium | **Labels:** infra, security
- **Description:** Move in-memory limiter in [auth-security.ts](#) to Redis when configured.
- **Acceptance criteria:** Documented in [PRODUCTION_READINESS.md](#) L43

COS-504 — Push notification channel (Expo / Web Push)

- **Priority:** Medium | **Labels:** notifications
- **Description:** `NotificationChannel.PUSH` exists in schema but unused; `EXPO_PUSH_ACCESS_TOKEN` in `.env.example` has no code path.
- **Acceptance criteria:**
 - Provider in [notification-provider.ts](#)
 - Mobile opt-in + device token registration endpoint
 - At least order-shipped push trigger

COS-505 — OpenTelemetry instrumentation

- **Priority:** Low | **Labels:** infra, observability
- **Description:** Env vars exist (`PLEROS_OTEL_ENABLED`); wire traces for API routes and order saga.
- **Acceptance criteria:** Spans visible in local OTEL collector ([infra/docker/otel-config.yaml](#))

Epic 6: ERP Roadmap (from gap analysis)

Per [ERP_FEATURE_GAP.md](#) L231 suggested order:

COS-601 — Purchase requisition + approval workflow

- **Priority:** Medium | **Labels:** purchasing, erp
- **Description:** Requisition entity, approval chain, convert-to-PO. Today: direct PO only (L109).
- **Acceptance criteria:** Admin UI + API; approver notification

COS-602 — Batch/lot recall workflow and compliance UI

- **Priority:** Medium | **Labels:** compliance, wms
- **Description:** `Batch.recalled` exists in [compliance/schema/prisma](#) L87 but no admin recall flow or webhook emit.
- **Acceptance criteria:**
 - Admin action: mark batch recalled → block allocation/ship
 - Emit `compliance.batch_recalled` webhook (ties to COS-501)
 - Customer/order impact report

COS-603 — Deeper sales tax engine (nexus, exemptions)

- **Priority:** Medium | **Labels:** finance, compliance
- **Description:** Extend [compliance-tax.ts](#) beyond static state rates.
- **Acceptance criteria:** Customer tax exemption certs; nexus jurisdiction config

COS-604 — Fine-grained RBAC per module

- **Priority:** Medium | **Labels:** security, admin
- **Description:** [permissions.ts](#) permission map exists but routes use coarse role guards only.
- **Acceptance criteria:** `assertPermission` used on sensitive routes; admin role editor UI

COS-605 — Report builder / export suite

- **Priority:** Medium | **Labels:** analytics, admin
- **Description:** KPI dashboard exists; no ad-hoc reports or CSV/Excel export suite.
- **Acceptance criteria:** Saved reports for orders, inventory, AR aging; CSV download

COS-606 — Postgres unified local dev path

- **Priority:** Low | **Labels:** infra, dx
- **Description:** [MISSING.md](#) L12 — local defaults to SQLite; document or default docker-compose Postgres dev flow.
- **Acceptance criteria:** `npm run dev` works with Postgres via `docker compose up db`

Epic 7: Enterprise (Phase 2+ — defer unless customer-driven)

From [ERP_FEATURE_GAP.md](#) L219–228 — create Linear issues but mark **Backlog / Enterprise**:

ID	Title
COS-701	Multi-currency and FX revaluation
COS-702	Multi-subsidiary consolidation
COS-703	Rebate and vendor allowance programs
COS-704	Fixed assets and depreciation
COS-705	Manufacturing / BOM / kitting
COS-706	HR and payroll module
COS-707	Advanced BI / demand forecasting beyond EWMA
COS-708	Omnichannel connectors (Shopify, Amazon)
COS-709	Managed X12/AS2 EDI hub (beyond JSON interchange)

Recommended sprint order

Sprint 1 — Close marketing gaps (highest user-visible risk): COS-201, COS-202, COS-401, COS-301, COS-302

Sprint 2 — Regulated vertical readiness: COS-101, COS-102, COS-103, COS-104

Sprint 3 — Platform reliability: COS-501, COS-502, COS-602, COS-503

Sprint 4 — PWA polish + field mobile: COS-303, COS-304, COS-402, COS-203, COS-204

Sprint 5 — ERP mid-market parity: COS-601, COS-603, COS-604, COS-605

Linear setup notes

- **Team suggestion:** Cosmos or Pleros product team
- **Labels:** compliance , pwa , mobile , wms , pos , integrations , erp , enterprise
- **Priority mapping:** High → P1, Medium → P2, Low/Enterprise → P3/Backlog
- **No Linear integration exists in repo** — tasks above are ready to bulk-create manually or via Linear CSV import / API
- **Prefix:** Replace COS- with your Linear team prefix (e.g. PLR-)

Key files to touch (cross-cutting)

Feature	Primary paths
Age verification	apps/web/prisma/tenant/schema.prisma , apps/web/lib/server/orders.ts , apps/client/src/pages/admin/pos/ , apps/client/src/pages/checkout/
Label scanner	New apps/client/src/components/mobile/barcode-scanner.tsx , apps/client/src/pages/m/warehouse/
PWA	apps/client/public/sw.js , apps/client/src/lib/offline-sync.ts , apps/client/src/lib/offline-queue.ts
Webhooks	apps/web/lib/server/webhooks.ts , order/compliance server modules
Recall	apps/web/prisma/compliance/schema.prisma , new admin compliance recall UI